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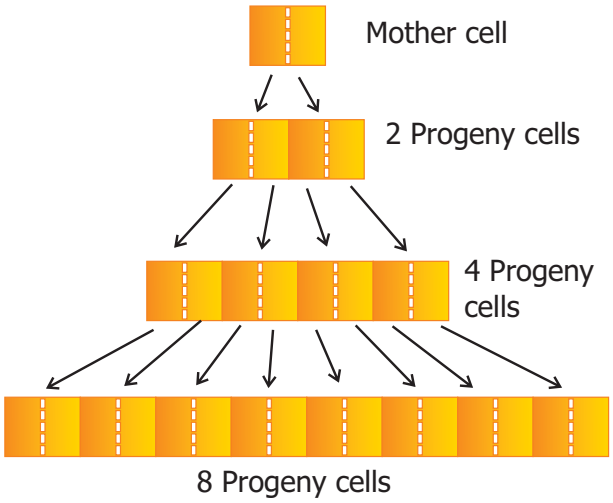


Figure 15.5: Geometric growth

$W_1 = W_0 e^{rt}$

W_1 = Final size (weight, height and number)

W_0 = Initial size at the beginning of the period

r = Growth rate

t = Time of growth

e = Base of the natural logarithms

Here '*r*' is the relative growth rate and also a measure of the ability of the plant to produce new plant material, referred to as efficiency index. Hence, the final size of W_1 depends on the initial size W_0.

iii. Arithmetic and Geometric Growth of Embryo

Plants often grow by a combination of arithmetic and geometric growth patterns. A young embryonic plant grows geometrically and cell division becomes restricted to certain cells at the tips of roots and shoots. After this point, growth is of the slower arithmetic type, but some of the new cells that are produced can develop into their mature condition and begin carrying out specialized types of metabolism (Figure 15. 6). Plants are thus a mixture of older, mature cells and young, dividing cells.

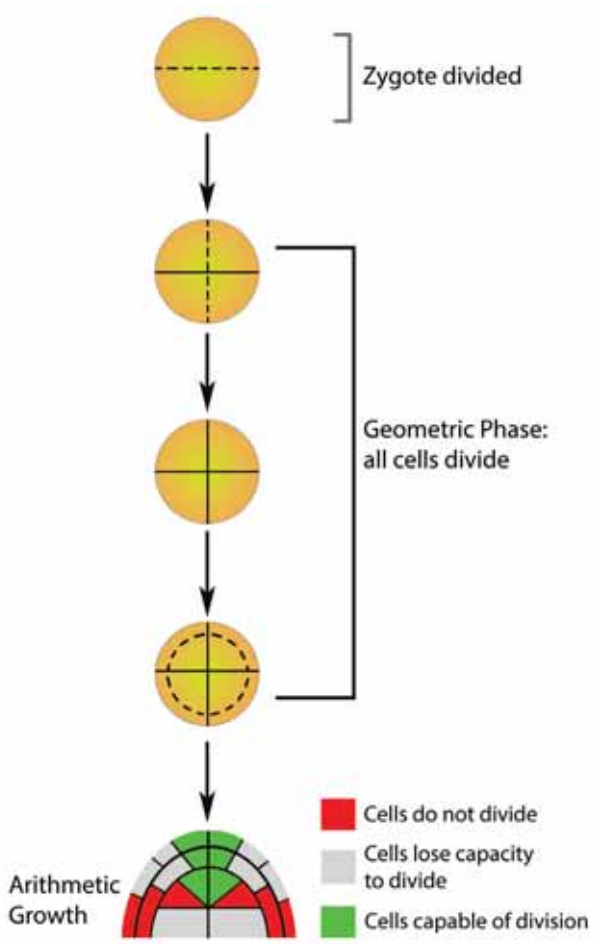


Figure 15.6: Arithmetic and geometric growth of embryo

Quantitative comparisons between the growth of living system can also be made in two ways and is explained in the table 1.

In figure 15.7, two leaves A and B are drawn at a particular time. Then $A^1$and B^1 are drawn after a given time. A and B = Area of leaves at a particular time. A^1 and B^1 = Area of leaves after a given time. (A^1-A) and (B^1-B) represents an absolute increase in area in the given time. Leaf A

Table 1: Comparison between absolute and relative growth rates

Absolute growth rate	Relative growth rate
Increase in total growth of two organs measured and compared per unit time is called absolute growth rate.	The growth of the given system per unit time expressed per unit initial parameter is called relative growth rate.